An Execution-Time CBF-QP Safety Filter on Frozen MAPPO Formation Policies: A Pre-Specified Paired Evaluation in Simulation

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DRAFT - not for distribution; authorship and framing pending advisor review

AI-assistance statement. The code, analyses and this draft were produced with substantial assistance from an AI coding assistant (Claude, Anthropic) under the author's direction. Every number in the text is generated by committed scripts from committed result files (`scripts/paper_numbers.py`, `paper/numbers.tex`). The author is responsible for the content.

Abstract

We evaluate whether an execution-time, decentralised control-barrier-function quadratic-programme (CBF-QP) safety filter, placed between frozen multi-agent PPO (MAPPO) formation policies and a simulated differential-drive team, is useful under criteria fixed before the test. Nine frozen actors (three training seeds, three checkpoints each) were evaluated on reserved, previously unused layouts with one, two and three obstacles (9,000 paired trials per density), with the filter off, the filter on at the gain $\alpha^\star = 10$ selected on validation layouts, and a brake comparator. The filter reduced the share of trials ending in collision or infeasibility at every density (from 5.4% to 0.1%, 16.6% to 0.8% and 48.0% to 3.9%). Under the pre-set criteria it was nevertheless *not shown useful* at any density: the non-inferiority criteria for goal completion (margin five percentage points) and for formation error over the common executed prefix (margin 0.01 m) were not met. Exploratory, post hoc analyses of the saved logs indicate that many filtered runs stalled in front of obstacles on layouts the unfiltered policy completed or collided on, and that in all 433 infeasible trials the row with the most negative zero-command slack lay within $4.3 \times 10^{-8}\,\mathrm{m}^2$ of its stand-off, consistent with numerical sensitivity at $\alpha\Delta = 1$. Claims cover these nine actors in this simulated scenario only.

1 Introduction

Reinforcement-learning policies for multi-robot formations are commonly trained without safety constraints [1], and a control-barrier-function (CBF) safety filter [2] can be added at execution time without retraining. Whether such a filter is *useful*, i.e. reduces failures without an unacceptable loss of task completion or formation quality, is an empirical question for a given policy family. This draft reports a pre-specified paired evaluation of one such filter, a decentralised sampled-data CBF-QP, on frozen MAPPO [22] actors in a VMAS [23] differential-drive formation-navigation scenario, followed by clearly labelled exploratory analyses of why the pre-set criteria were not met.

Contributions. Studies of safe multi-agent learning mostly add safety during training or learn it together with the policy. Among those that act at execution, one evaluates with and without its filter a policy that was trained with the filter, one shows the unfiltered policy only qualitatively with a centralised filter on double integrators, and one adds a learned correction to a frozen policy without stating that its comparison is paired (Section 2, Table 1). This draft adds: (i) a paired evaluation, with criteria and margins fixed before the test and reserved test layouts, of a decentralised execution-time CBF-QP filter on a family of nine frozen MAPPO actors that never saw the filter, together with a brake comparator on the same layouts; (ii) a confirmatory result for these actors: failures fell at every density, but non-inferiority in goal completion and formation error was not established, with outcomes, failure types and per-seed variation reported in full; (iii) exploratory, clearly labelled analyses of where the cost arose (stalls in front of obstacles, small unfiltered clearances in the lost completions, infeasibility at the stand-off when $\alpha\Delta = 1$) and documented analytic predictions that the development data matched; and (iv) the code, saved per-step logs of every final-test trial and scripts that regenerate every number, so the analyses can be repeated without re-running trials.

Research question. Does an execution-time CBF-QP filter, applied to frozen MAPPO formation policies trained without it, reduce collisions, and at what cost in formation error, goal completion and deadlock?

Motivating hypotheses (Stage 2). The study was motivated by three hypotheses stated in an earlier (Stage 2) submission, quoted here: "H1: collisions fall. H2: formation error and completion time rise, more so in denser layouts. H3: most failures under filtering come from deadlock or infeasible steps rather than slower progress." Stage 2 defined deadlock as "centroid progress below a set distance over a set window" (the protocol's stalled flag). These hypotheses motivate the design; they do not extend the locked confirmatory claims, which are the three criteria of Section 5. Section 8 maps each hypothesis to the evidence and states which evidence is confirmatory and which is exploratory.

2 Related work

Safe learning in robotics is surveyed in [1] and control barrier functions in [2]. Multi-agent approaches differ in when safety enters. Some apply a CBF shield during training [6] or learn barrier certificates together with the policy [7, 10, 8]; some filter during both training and execution [9, 5]; one learns a safety correction on top of a frozen MAPPO policy [4]; and one filters at execution a policy trained without the filter, using a centralised QP for double-integrator agents [12]. Table 1 summarises them. Here *paired on/off* means that the same frozen policy is evaluated with and without the filter on matched layouts; a comparison between separately trained policies (a training ablation) does not count as paired in this sense. [5] does report a paired test in this sense, but for a policy trained with its filter, which is then deactivated; the present study evaluates policies that never saw the filter. The classifications were checked against the papers on 26 October 2026: against the full text for [6, 9, 5, 4, 12] and against the abstract only for [7, 10, 8], where "not reported in abstract" marks what could not be checked.

Sampled-data CBF margins that account for inter-sample behaviour are studied in [13, 14, 15, 16]; these margins are input-independent bounds over the input set and do not depend on the applied command. Decentralised equal-split ("half-responsibility") constraints go back to [3, 17], and stopping as a feasibility-preserving backup is used in [18]. The offset (barrier) point for unicycles is used in [19]. Hidden assumptions in CBF safety claims are discussed in [20]. Input-dependent

Table 1: Related multi-agent studies. "Paired on/off" = the same frozen policy evaluated with and without the filter on matched layouts. Checked against the full text (F) or the abstract only (A) on 26 October 2026.

Study	When safety enters	Paired on/off	Outcomes reported	Checked
Cai et al. [6]	CBF shield during training (also claimed for deployment)	No: separately trained agents	collisions, training reward	F
Qin et al. [7]	certificates learned with the policy; test-time refinement	not reported in abstract	safety, task completion	A
GCBF+ [10]	graph CBF and distributed policy learned together	not reported in abstract	safety, goal reaching	A
DGPPO [8]	graph CBF learned simultaneously with the policy	not reported in abstract	task performance, safety rate	A
Choi et al. [9]	CBVF filter in the second half of training and at execution	No: separately trained variants	goal reach, near collisions, conflicts, travel time	F
Deng et al. [4]	learned correction added to a frozen MAPPO policy	base policy reported; pairing not stated	return, arrival rate, violation rate	F
Dawood et al. [5]	MPC filter in training and execution	Yes, for a policy trained with the filter	goals, timeouts, collisions, time, formation error	F
Vinod et al. [12]	centralised QP filter at execution; policy trained without it	same policy and start states, shown qualitatively	completion, deadlock, loitering	F
This study	decentralised CBF-QP at execution; policies trained without it	Yes: 9,000 matched layouts per density	collisions, infeasibility, completion, formation, stalls	–

inter-sample bounds used for trigger timing appear in [21]. Interval estimates and per-seed reporting for RL evaluation are argued for in [11].

3 System and filter

Robots and task. Three differential-drive robots with axle-centre pose $x = (x, y, \theta)$, body disc of radius $R = 0.10\,\mathrm{m}$, and command $u = (v, \omega)$ in the box $\mathcal{U} = \{-0.5 \le v \le 0.5\ \mathrm{m/s},\ |\omega| \le 1\ \mathrm{rad/s}\}$ follow the unicycle model

$$\dot{x} = v\cos\theta, \qquad \dot{y} = v\sin\theta, \qquad \dot{\theta} = \omega, \tag{1}$$

with commands held over each control period $\Delta = 0.1\,\mathrm{s}$ (zero-order hold). The team moves a triangular formation (slot radius 0.30 m) through a field of K circular obstacles of radius $R_o = 0.20\,\mathrm{m}$ to a goal within a horizon of 400 periods. GOAL requires the centroid within 0.10 m of the goal and mean slot error within 0.10 m; COLLISION is scored on the swept path and ends the trial; an infeasible QP ends the trial as INFEASIBLE before any motion; otherwise the trial ends as TIMEOUT.

Offset point and barriers. The filter constrains the point $p = (x+\ell\cos\theta,\ y+\ell\sin\theta)$, $\ell = 0.05\,\mathrm{m}$, for which

$$\dot{p} = J(\theta)\,u, \qquad J(\theta) = \begin{bmatrix} \cos\theta & -\ell\sin\theta \\ \sin\theta & \ell\cos\theta \end{bmatrix}, \qquad \det J = \ell, \tag{2}$$

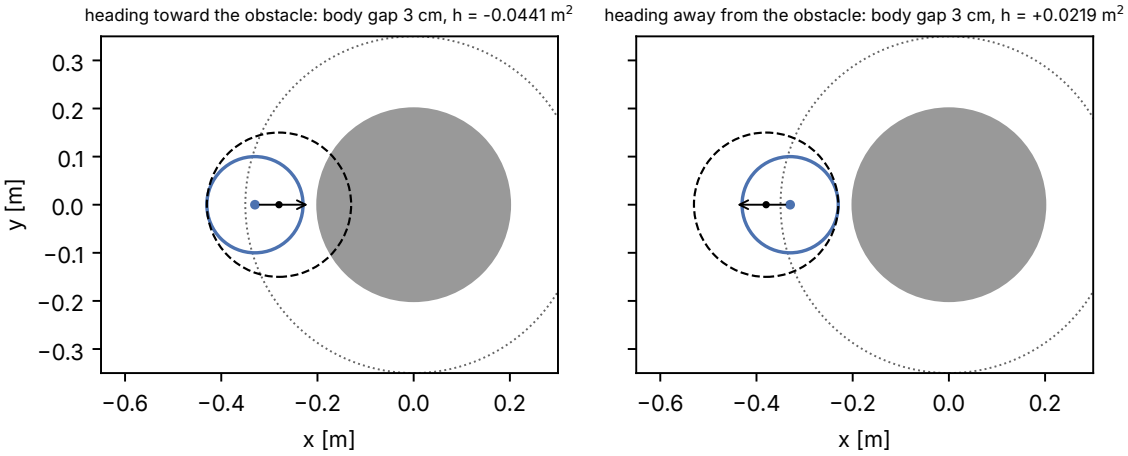


Figure 1: Analytic schematic (no data). Body disc (solid), axle centre, offset point p and cover disc of radius $R+\ell$ (dashed) next to an obstacle (grey) whose barrier boundary $\|p-c\| = D_o$ is dotted. With the same 3 cm body clearance, the cover barrier is negative when the robot heads toward the obstacle and positive when it heads away.

so every barrier below has relative degree one in (v, ω). The barriers are

$$h_o = \|p-c\|^2 - D_o^2, \quad D_o = R+\ell+R_o; \qquad h_{ij} = \|p_i - p_j\|^2 - D_{rr}^2, \quad D_{rr} = 2(R+\ell). \tag{3}$$

Because $\|x-p\| = \ell$, the body disc lies inside the *cover disc* of radius $R+\ell$ around p, so $h \geq 0$ implies physical separation. The converse does not hold: depending on the heading, a robot whose body is a few centimetres from an obstacle can have $h_o < 0$ (Fig. 1).

Sampled-data condition. The implemented filter uses the following lemma with a two-sided bound on $\ddot{h}$; only the lower bound is needed.

Lemma 1 (one-sided form). Let u be held on $[t_k, t_k+\Delta]$, $\ddot{h} \geq -M$ on that interval with $M \geq 0$, $\alpha\Delta \leq 1$ and $h(t_k) \geq 0$. If $\dot{h}(t_k) + \alpha h(t_k) \geq \delta = M\Delta/2$, then $h(t_k+s) \geq (1-\alpha s)\, h(t_k) \geq 0$ for all $s \in [0, \Delta]$.

This is the second-order approach of [14] (Theorem 3, stated there with the opposite sign convention for h; $\gamma \leq 1$ corresponds to $\alpha\Delta \leq 1$). With $V_p = \sqrt{\bar{v}^2 + \ell^2\bar{\omega}^2}$ and $A_p = \bar{\omega}V_p$, the implemented margins use the two-sided bounds of the design document,

$$\delta_o(d) = \tfrac{\Delta}{2}\left[2V_p^2 + 2(d + V_p\Delta)A_p\right], \qquad \delta_{ij}(d) = \tfrac{\Delta}{2}\left[8V_p^2 + 4(d + 2V_p\Delta)A_p\right], \tag{4}$$

where d is the current barrier-point distance. At $\alpha\Delta = 1$ the lemma gives $h(t_k+\Delta) \geq 0$ but no strictly positive endpoint bound (Section 8).

Equal split and the per-robot QP. With $g_o = 2J(\theta_i)^\top(p_i - c)$ and $g_{ij} = 2J(\theta_i)^\top(p_i - p_j)$, robot i solves at every period

$$\begin{aligned} \min_{u \in \mathcal{U}} \ & \tfrac{1}{2}\|u - u_{\text{nom}}\|^2 \\ \text{s.t.} \ & g_o^\top u \ \geq -\alpha h_o + \delta_o(d) \qquad \text{obstacles with } d \leq 1.0\,\text{m}, \\ & g_{ij}^\top u \ \geq -\tfrac{\alpha}{2} h_{ij} + \tfrac{1}{2}\delta_{ij}(d) \quad \text{neighbours with } d \leq 1.5\,\text{m}, \end{aligned} \tag{5}$$

Table 2: Filter-off baseline on validation layouts (200 per cell): GOAL / COLLISION / TIMEOUT (%) per actor and obstacle count K (descriptive context; $K = 4$ was an intermediate diagnostic).

actor	$K = 1$	$K = 2$	$K = 3$	$K = 4$	$K = 5$	$K = 8$
s0_early	82/5/13	52/26/22	16/63/22	6/84/10	0/98/2	0/100/0
s0_middle	64/3/33	48/24/28	14/58/28	7/80/12	0/98/2	0/100/0
s0_late	56/4/39	50/24/26	16/58/27	5/80/15	0/98/2	0/100/0
s1_early	17/12/72	56/21/23	25/56/20	12/84/4	0/97/2	0/100/0
s1_middle	42/10/48	75/13/12	35/48/17	18/76/6	1/93/6	0/100/0
s1_late	44/5/50	78/10/12	25/48/26	20/74/6	4/90/6	0/100/0
s2_early	24/4/72	72/11/16	58/28/14	19/70/11	2/96/2	0/100/0
s2_middle	53/5/42	75/14/11	50/31/19	5/82/12	0/96/3	0/100/0
s2_late	76/5/18	80/16/5	49/34/18	18/78/4	0/98/2	0/100/0

where u_{nom} is the frozen actor's deterministic command. Each robot takes half of every pair condition (equal split); the two halves sum to the pair condition of Lemma 1 when both robots use the same state. The two-variable QP is solved by enumerating its candidate optima (the unconstrained optimum, the projection onto each constraint line and every pairwise vertex) and keeping the closest feasible one; this is implemented in floating-point arithmetic with a feasibility tolerance of 10^{-9} on normalised rows. There is no fallback, so an infeasible QP ends the trial. The B-match comparator applies $(0, \omega_{\text{nom}})$ whenever the clipped nominal command violates one of these rows at the same gain; it is not a certified stop (the offset point rotates).

4 Policy and training

Observation and reward. Each robot observes, in its body frame: its slot error relative to the centroid, the centroid-to-goal vector (norm-clipped at 2.0 m), teammates' relative positions and headings, the eight nearest obstacles within 2.0 m (relative centre, surface gap, present flag) and an agent one-hot. The per-period reward is $10\cdot$(centroid progress toward the goal) $-1\cdot$(mean slot error) $-0.01\cdot$(control effort) $-10\cdot$[collision] $-0.2 \cdot \sum_{\text{pairs}} \max(0, 1 - \text{clearance}/0.15\,\text{m})$; a collision ends the episode and a timeout is bootstrapped. The filter is never used in training.

MAPPO and curriculum. A compact PyTorch implementation of MAPPO [22] (shared Gaussian actor with tanh-bounded mean, centralised critic, GAE with $\gamma = 0.99$, $\lambda = 0.95$, clip 0.2, 10 epochs, 16 minibatches, learning rates 3×10^{-4} actor and 10^{-3} critic, 384 parallel environments of 64 steps) was trained with three seeds through a curriculum $K = 0 \to 1 \to 2 \to 3 \to (2, 5, 8)$ with stage budgets of 1, 2, 4, 4 and 8 million environment steps. The implementation was sanity-checked only on the stock VMAS navigation task, where the deterministic policy reached 99.0% success on 200 fixed episodes (uniform random: 0.0%).

Checkpoints and gate history. The nine actors are the early, middle and late checkpoints of the final curriculum stage of each seed, fixed by an iteration rule (first saved checkpoint with $r \geq \lceil n/3 \rceil$, $r \geq \lceil 2n/3 \rceil$, and $r = n$, with $n = 326$ final-stage iterations). The original pre-registered filter-off gate (pooled GOAL $\geq 50\%$ over $K = 2, 5, 8$) was not met by any checkpoint: the best pooled GOAL rate was 11.0% (first configuration) and 28.7% (curriculum). $K = 5, 8$ were therefore moved to stress cells and $K = 1, 2, 3$ became the primary densities. Table 2 shows the filter-off baseline of the nine actors on validation layouts.

5 Experimental design

The protocol was locked by the author before any filter-on data with the frozen MAPPO actors existed; the margins and decisions were chosen without advisor review (Section 12). Filter-on results did exist before the lock in two other forms: engineering stress tests of the filter with random and adversarial stand-in controllers (4,200 paired trials) and paired demonstration runs with an obstacle-blind stand-in controller. Neither used the frozen MAPPO actors. The frozen-MAPPO development sweep on validation layouts followed the lock, and the reserved final evaluation followed the gain selection.

The gain was selected on existing validation layouts from $\alpha \in \{0.5, 1, 2, 5, 10\}\,\mathrm{s}^{-1}$ by the highest pooled filter-on GOAL count over the nine actors and $K = 1, 2, 3$ (ties: fewer INFEASIBLE, then the larger gain). The final test used 1,000 reserved layouts per actor and density, each run with the filter off, the filter on at $\alpha^\star$ and B-match at $\alpha^\star$. Usefulness at a density required all of: (a) a one-sided exact McNemar reduction in COLLISION-or-INFEASIBLE; (b) GOAL loss (off minus on) with one-sided 98.33% upper bound at most 0.05; (c) the equal-actor-weighted mean increase in formation error over the common executed prefix with stratified bootstrap 98.33% upper bound at most 0.01 m; each one-sided at 0.05/3. For (a) and (b) the paired difference in proportions has a two-sided 96.67% Newcombe score interval [24] (Wilson score limits combined through the estimated correlation $\hat{\phi}$), whose appropriate limit gives the one-sided 98.33% bound. The locked protocol names this "method 10", but the analysis code, committed before any frozen-MAPPO filter-on data, uses the uncorrected $\hat{\phi}$, which is Newcombe's method 8; method 10 applies a continuity correction to $\hat{\phi}$. This was found after the final test. Recomputing every bound for (a) and (b) with method 10 changes it by at most 0.0008 percentage points and changes no criterion (`scripts/newcombe_method10_check.py`); the reported numbers are those of the locked code. Per-seed and per-actor results, episode length, reached-goal-region (centroid within 0.30 m of the goal before any collision or infeasibility), final centroid-to-goal distance, TIMEOUT and the stalled flag were pre-declared as secondary and descriptive.

Formation endpoint (criterion (c)). For a trial, let T be the number of control periods actually executed: a period that ends in COLLISION counts, while an attempted period that an infeasible QP prevented does not. Let $x_r(t)$ be robot r's position at the post-step state after period t, $\bar{x}(t)$ the team centroid and s_r its slot offset; the formation error is

$$e(t) = \tfrac{1}{3}\textstyle\sum_{r=1}^{3} \big\| x_r(t) - \bar{x}(t) - s_r \big\|, \qquad t = 1, \ldots, T, \tag{6}$$

so the initial state and the frozen samples after termination are excluded. For pair i with executed counts T_{off} and T_{on}, the common executed prefix has length $L_i = \min(T_{\mathrm{off}}, T_{\mathrm{on}})$ and, for $L_i > 0$,

$$d_i = \frac{1}{L_i} \sum_{t=1}^{L_i} \big[e_{\mathrm{on}}(t) - e_{\mathrm{off}}(t) \big]. \tag{7}$$

Pairs with $L_i = 0$ (an arm infeasible at its first period) have no value: they are excluded from (c), never counted as zero, and remain failures in (a) and non-GOAL outcomes in (b). With $\mathcal{E}_p$ the eligible pairs of actor p, the estimand is the equal-actor mean

$$\bar{d} = \frac{1}{9} \sum_{p=1}^{9} \frac{1}{|\mathcal{E}_p|} \sum_{i \in \mathcal{E}_p} d_i. \tag{8}$$

Table 3: Confirmatory results, filter on ($\alpha^\star = 10$) versus filter off, nine actors pooled, 9,000 pairs per density. Differences in percentage points (pp); one-sided bounds at 98.33%; 95% intervals descriptive.

	$K = 1$	$K = 2$	$K = 3$
COLLISION-or-INFEASIBLE, off → on (%)	5.4 → 0.1	16.6 → 0.8	48.0 → 3.9
(a) difference on−off [pp]	−5.30	−15.72	−44.08
98.33% upper; 95% CI	−4.82; [−5.78, −4.85]	−14.92; [−16.49, −14.98]	−42.95; [−45.11, −43.04]
one-sided exact p; criterion	2.6×10^{-144}; met	$< 10^{-300}$; met	$< 10^{-300}$; met
GOAL off / on (%)	54.7 / 49.9	66.6 / 56.8	32.0 / 21.5
(b) GOAL loss off−on [pp]	4.81	9.81	10.56
98.33% upper (margin 5); criterion	5.31; not met	10.59; not met	11.33; not met
(c) formation increase [m]	0.0117	0.0202	0.0278
bootstrap 98.33% upper (margin 0.01)	0.0127	0.0213	0.0291
normal cross-check; eligible / excluded	0.0126; 9,000/ 0	0.0213; 9,000/ 0	0.0291; 9,000/ 0
criterion	not met	not met	not met
Verdict	not shown useful	not shown useful	not shown useful

Its upper bound is a stratified paired percentile bootstrap: pairs are resampled with replacement within each actor, $\bar{d}$ is recomputed with the same equal weights, $B = 20{,}000$ replicates are drawn with `numpy.random.default_rng(20261026)`, and the bound is the $1 - 0.05/3$ quantile of the replicates (`numpy.quantile`, method `linear`). A normal bound from the per-actor variances is reported as a cross-check. If any actor has fewer than 2 eligible pairs, criterion (c) is inconclusive and the weights are not renormalised. In the final test every pair was eligible (Table 3).

6 Results (confirmatory)

Gain selection (development data). Pooled filter-on GOAL counts over 5,400 validation trials were 0, 255, 1,184, 1,936 and 2,224 for $\alpha = 0.5, 1, 2, 5, 10$, with 1,251, 4, 11, 61 and 129 INFEASIBLE outcomes, so $\alpha^\star = 10$ (Fig. 2).

Table 3 gives the confirmatory results. Criterion (a) was met at every density; (b) and (c) were met at none, so the filter was *not shown useful under the pre-set criteria* at $K = 1, 2, 3$. With the filter on there were 0, 0 and 0 collisions and 6, 75 and 352 INFEASIBLE trials; TIMEOUT rose from 39.9%, 16.9% and 20.0% to 50.1%, 42.4% and 74.6% (Fig. 3). B-match had 0, 0 and 0 collisions and 0, 0 and 0 INFEASIBLE outcomes, and reached GOAL in 48.8%, 53.5% and 19.3% of trials; the filter reached GOAL 1.0, 3.2 and 2.2 percentage points more often (descriptive).

Reached goal region (pre-declared secondary, descriptive). The share of trials whose centroid came within 0.30 m of the goal before any collision or infeasibility fell from 94.6% to 87.1%, 83.2% to 69.5% and 51.3% to 35.0% at $K = 1, 2, 3$ (paired differences −7.5 [−8.1, −7.0], −13.7 [−14.4, −12.9] and −16.3 [−17.2, −15.5] percentage points, 95% intervals). These losses are at least as large as the GOAL losses, so the lost completions were mostly runs that never approached the goal, not runs that approached it and missed the strict 0.10 m parking tolerance (descriptive comparison of point estimates). The remaining pre-declared secondary results (final centroid-to-goal distance, formation error over the full trajectory in pairs where both arms reached GOAL, intervention and brake rates, sampled barrier minima, and filter-on versus B-match comparisons) are listed in `results/final_test/FINAL_TEST.md` and `endpoints.csv`.

Table 4: Locked endpoints per training seed (pre-declared secondary, descriptive; three actors and 3,000 pairs per row; 95% intervals; no tests).

seed	K	COLLISION-or-INFEASIBLE off → on (%)	(a) difference [pp]	(b) GOAL loss [pp]	(c) formation increase [m]
0	1	3.9 → 0.0	−3.9 [−4.7, −3.3]	2.9 [2.3, 3.5]	0.0047 [0.0040, 0.0054]
1	1	8.5 → 0.2	−8.3 [−9.4, −7.4]	7.5 [6.5, 8.5]	0.0252 [0.0228, 0.0276]
2	1	3.7 → 0.0	−3.7 [−4.4, −3.0]	4.0 [3.3, 4.8]	0.0052 [0.0043, 0.0061]
0	2	24.5 → 1.3	−23.2 [−24.7, −21.7]	6.4 [5.3, 7.5]	0.0222 [0.0204, 0.0241]
1	2	14.0 → 0.5	−13.5 [−14.8, −12.3]	9.0 [7.8, 10.3]	0.0201 [0.0182, 0.0219]
2	2	11.1 → 0.7	−10.5 [−11.6, −9.4]	14.0 [12.6, 15.3]	0.0183 [0.0167, 0.0199]
0	3	62.4 → 7.0	−55.4 [−57.2, −53.6]	7.9 [6.9, 8.9]	0.0276 [0.0254, 0.0298]
1	3	51.1 → 2.5	−48.6 [−50.4, −46.8]	6.7 [5.6, 7.8]	0.0261 [0.0239, 0.0283]
2	3	30.5 → 2.3	−28.3 [−29.9, −26.7]	17.1 [15.5, 18.6]	0.0297 [0.0277, 0.0317]

Table 5: Time to goal in joint-success pairs (periods of 0.1 s; test arm minus filter off; pre-declared episode-length endpoint, exploratory joint-success analysis).

arm	K	pairs	fewest per actor	equal-actor mean	median	HL	slower (%)	Wilcoxon p
filter on	1	4,472	199	0.30	0.0	0.0	9.0	2.6×10^{-57}
filter on	2	4,968	406	0.68	0.0	0.0	23.9	1.9×10^{-89}
filter on	3	1,815	65	2.09	0.0	0.5	41.3	5.1×10^{-90}
B-match	1	4,385	199	0.47	0.0	0.0	10.1	1.3×10^{-59}
B-match	2	4,671	370	−0.24	0.0	0.0	27.3	5.5×10^{-92}
B-match	3	1,629	53	2.62	0.0	1.0	44.3	3.6×10^{-90}

Per seed and per actor (pre-declared secondary, descriptive). Table 4 gives the three locked endpoints per training seed with 95% intervals; no test or verdict is made per seed, and a pooled result does not establish usefulness, or its absence, for each seed or actor. Per-actor GOAL losses vary widely (Fig. 4).

Time to goal (pre-declared episode-length endpoint; exploratory, post hoc joint-success analysis). In pairs where both arms reached GOAL, Table 5 compares executed periods to GOAL (test arm minus filter off; positive = slower). The equal-actor mean weights the nine actors equally; the median, Hodges–Lehmann estimate and Wilcoxon test are pair-weighted. Conditioning on joint success selects the easier layouts and does not estimate the overall completion-time cost.

Validation versus final test (exploratory, post hoc). Table 6 sets the development-sweep estimates at $\alpha = 10$ (validation layouts, used for gain selection) beside the final-test estimates.

Stress cells (exploratory, post hoc, descriptive). The development sweep also ran the diagnostic and stress densities $K = 4, 5, 8$ on validation layouts (Table 7). They were not part of the final test and did not affect gain selection.

7 Exploratory mechanism analysis (exploratory, post hoc)

Everything in this section is exploratory and post hoc: categories, thresholds, groups and the example were chosen after the confirmatory results were known, the analyses read only saved logs, and they support no causal or confirmatory claim.

Table 6: Estimates at $\alpha = 10$ on validation layouts (development data) and on the reserved final test (exploratory, post hoc comparison; 95% intervals).

K	data	pairs	fail off → on (%)	GOAL off → on (%)	GOAL loss [pp]	formation increase [m]
1	validation	1,800	5.8 → 0.0	51.1 → 47.7	3.4 [2.5, 4.3]	0.0104 [0.0085, 0.0125]
1	final test	9,000	5.4 → 0.1	54.7 → 49.9	4.8 [4.4, 5.3]	0.0117 [0.0108, 0.0126]
2	validation	1,800	17.6 → 1.6	65.1 → 53.3	11.8 [10.1, 13.5]	0.0257 [0.0230, 0.0284]
2	final test	9,000	16.6 → 0.8	66.6 → 56.8	9.8 [9.1, 10.5]	0.0202 [0.0192, 0.0212]
3	validation	1,800	46.9 → 5.6	31.9 → 22.5	9.4 [7.8, 10.9]	0.0290 [0.0262, 0.0320]
3	final test	9,000	48.0 → 3.9	32.0 → 21.5	10.6 [9.8, 11.3]	0.0278 [0.0266, 0.0290]

Table 7: Stress cells on validation layouts at $\alpha = 10$ (outcome shares in %; exploratory, post hoc, descriptive; not part of the final test).

K	arm	trials	GOAL	COLLISION	INFEASIBLE	TIMEOUT
4	filter off	1,800	12.1	78.7	0.0	9.3
4	filter on (alpha = 10)	1,800	5.3	0.0	8.3	86.3
4	B-match (alpha = 10)	1,800	4.6	0.0	0.0	95.4
5	filter off	1,800	0.9	95.9	0.0	3.2
5	filter on (alpha = 10)	1,800	0.3	0.0	7.3	92.3
5	B-match (alpha = 10)	1,800	0.2	0.0	0.0	99.8
8	filter off	1,800	0.0	100.0	0.0	0.0
8	filter on (alpha = 10)	1,800	0.0	0.0	7.1	92.9
8	B-match (alpha = 10)	1,800	0.0	0.0	0.0	100.0

Stall classes. Of 15,038 filter-on TIMEOUTs, 9,211 (61%) were classified as filter-induced stalls (the clipped nominal command violated an enabled barrier row in at least half of the last 50 periods), 9,143 of them involving obstacle rows; on the same layouts the unfiltered run collided in 5,699 and reached GOAL in 2,274 cases. Another 5,670 (38%) were near-goal parking stalls, 5,362 of them on layouts where the unfiltered run also timed out.

Groups A and B. Group A (2,274 pairs: unfiltered GOAL, filtered filter-induced stall; 2,259 obstacle-labelled, 15 neighbour-only) and group B (11,255 pairs: both GOAL) were selected by observed outcome. The unfiltered runs of group A came closer to obstacles (median sampled minimum surface clearance 3.3 cm, group A only) than those of group B (14.4 cm; Fig. 5). Their applied commands violated at least one filter obstacle row in 89% (A) versus 4% (B) of runs, and a robot-pair row in 85% versus 36%.

Example (exploratory, post hoc). Fig. 6 shows one group-A pair, chosen by a rule fixed before plotting: the pair whose unfiltered minimum sampled obstacle clearance is closest to the unrounded group-A median (3.30 cm), ties by actor index, density and trial. In exact decimal comparison 2 candidates were equally close; the actor-index tie-break selected actor s1_late, $K = 2$, trial 348 (clearance 3.30 cm). The unfiltered run reached GOAL after 83 periods with its closest approach at period 21; the filtered run's centroid stalled from period 78. At that instant, robot 0 (the robot with the smallest obstacle cover barrier, $h = 0.0046\,\mathrm{m}^2$, body clearance 9.5 cm) is shown with its cover disc. This is a geometric selection: it does not show that this robot or constraint caused the stall, and a centroid stall does not mean that every robot stopped.

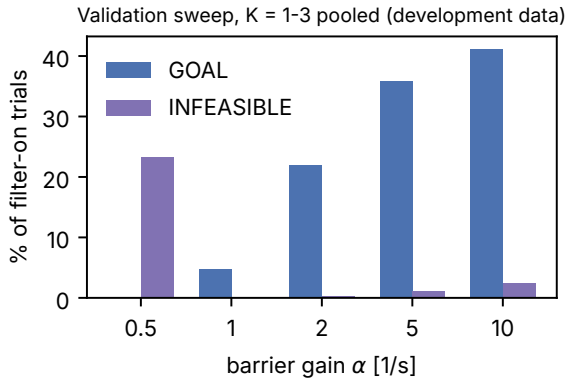


Figure 2: Validation sweep (development data): filter-on GOAL and INFEASIBLE share per gain, $K = 1$–3 pooled.

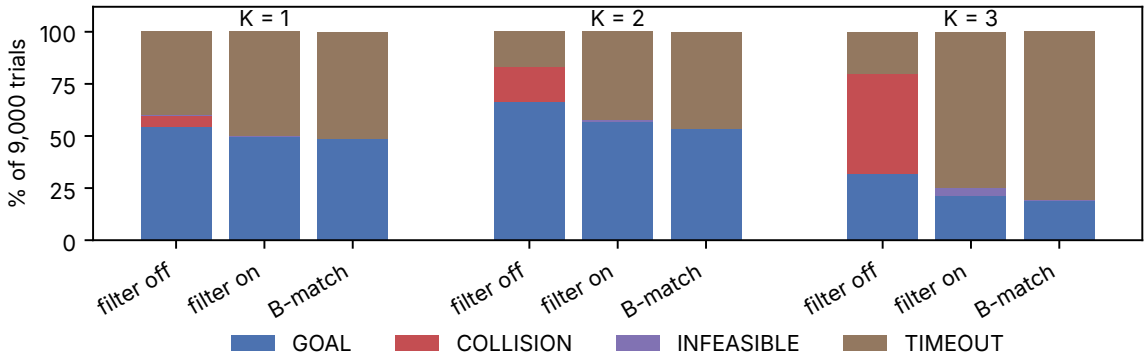


Figure 3: Final-test outcome shares per density for filter off, filter on ($\alpha^\star$) and B-match.

Clearance thresholds. Of all trials, 51.1% reached GOAL unfiltered, and 43.1% did so while keeping a sampled minimum obstacle clearance of at least 5 cm, against a filter-on GOAL rate of 42.7% on the same layouts (Fig. 5). This describes how close the unfiltered runs came to obstacles; it does not measure what a buffer or any filter would cost. Among unfiltered GOAL trials the descriptive AUC of the unfiltered minimum obstacle clearance for "filtered run did not reach GOAL" was 0.96.

Infeasibility. All 433 INFEASIBLE trials were reproduced (433 of 433) by the unchanged filter code at the saved states in an earlier audit. An infeasible QP has no optimal active set, so the audit examined, for each infeasible robot, the row with the most negative zero-command slack. In every case that row's barrier value was within $4.3 \times 10^{-8}\,\mathrm{m}^2$ of its stand-off $h^\star = \delta/\alpha$ and the zero command violated it by between -8.4×10^{-9} and -4.9×10^{-6} (normalised); 410 of 433 met a pre-declared near-boundary diagnostic. This is consistent with numerical sensitivity near the boundary, not proof of the cause.

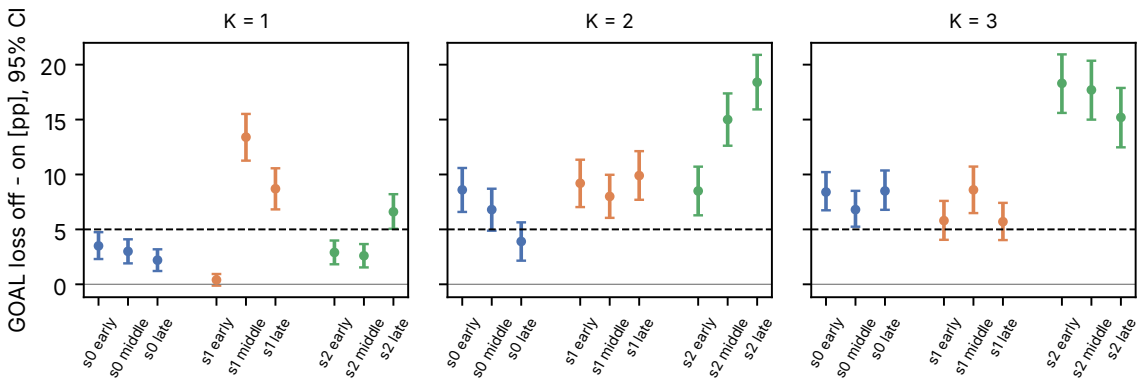


Figure 4: Per-actor GOAL loss (off minus on) with descriptive 95% Newcombe intervals, grouped by training seed; dashed line: the 5 pp margin of the pooled criterion (b).

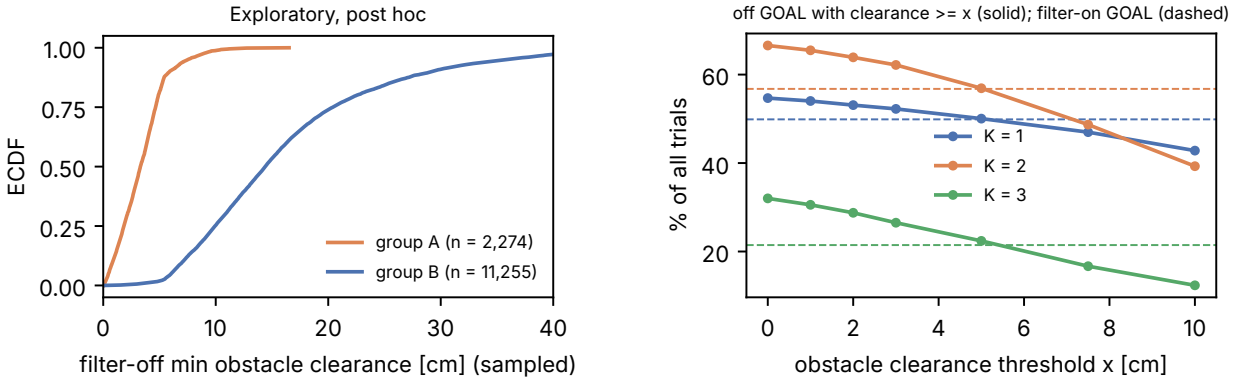


Figure 5: Exploratory, post hoc. Left (F4): unfiltered minimum obstacle clearance (sampled), group A vs group B. Right (F5): share of all trials that reached GOAL unfiltered with clearance at least x, per density, with the filter-on GOAL rate (dashed).

Transit. During formation transit the equal-split cap on a robot's closing speed toward a neighbour had median 0.72–0.78 m/s, above the largest barrier-point speed 0.50 m/s, and bound in 1–8% of moving closing periods.

8 Discussion

Hypotheses and evidence. *H1 (collisions fall).* Supported for these actors: with the filter on there were 0, 0 and 0 collisions against 483, 1,490 and 4,319 with the filter off (descriptive counts). The confirmatory endpoint is the composite COLLISION-or-INFEASIBLE, whose reduction met criterion (a) at every density (Table 3); collision counts alone were not a confirmatory endpoint. *H2 (formation error and completion time rise, more so in denser layouts).* The common-prefix formation increase was 0.0117, 0.0202 and 0.0278 m at $K = 1, 2, 3$. These point estimates exceed the 0.01 m margin, and criterion (c), which tests non-inferiority, was not met at any density: non-inferiority was not established (confirmatory). This is not the same as a confirmatory test showing that the increase exceeds the margin. The estimates increase with density, but no density trend

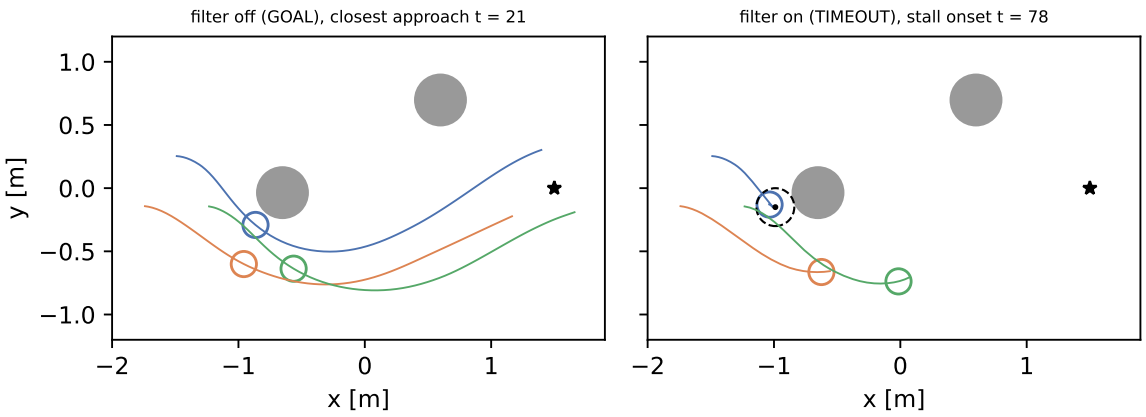


Figure 6: Exploratory, post hoc example. Selection rule (fixed before plotting): among group-A pairs, the pair whose filter-off minimum sampled obstacle clearance is closest to the unrounded group-A median; ties by actor index, density, then trial. Left: filter off, bodies at the closest-approach instant. Right: filter on, bodies at stall onset (first state from which the centroid stays within 0.05 m of its final position), with the cover disc (dashed, radius $R + \ell$) of the robot with the smallest obstacle barrier at that instant (ties by robot index). Star: goal. Geometric selection only; not evidence of cause.

was tested, so the "more so in denser layouts" part is not established. Completion fell (GOAL loss, criterion (b)); in joint-success pairs the time to goal changed little (equal-actor mean 0.30, 0.68 and 2.09 periods; Table 5), and this conditional analysis does not estimate the completion-time cost. *H3 (failures under filtering come from deadlock or infeasible steps rather than slower progress).* Of the 15,471 filter-on trials that did not reach GOAL, 99.8% ended INFEASIBLE or as TIMEOUT with the protocol's stalled flag, and 25 were TIMEOUTs still progressing at the horizon (stalled flag pre-declared as secondary, descriptive). The split of stalls into filter-induced and parking stalls is exploratory (Section 7).

Collisions versus stalls. With the filter on there were no collisions (0, 0 and 0 at $K = 1, 2, 3$; the remaining failures were INFEASIBLE; Table 3); in the exploratory classification most filtered timeouts were stalls with active obstacle rows, mostly on layouts where the unfiltered run collided or reached GOAL. We read this as a description of where the lost completions occurred, not as a causal attribution. The completions lost in group A involved unfiltered runs that passed close to obstacles (median 3.3 cm, group A only; Fig. 5). The filter reached GOAL 1.0–3.2 percentage points more often than B-match on the same layouts, with more INFEASIBLE outcomes.

Gain at the lemma boundary. The selected gain gives $\alpha^\star \Delta = 1$. At $\alpha^\star \Delta = 1$, the lemma provides no strictly positive endpoint lower bound from the initial barrier alone. Numerical robustness requires separate analysis. The infeasibility audit found the recorded infeasible states consistent with numerical sensitivity near this boundary.

Predictions. A methodology report written before the validation sweep, but committed to the repository only after the final test (`docs/methodology/README.md`), predicted that the selected

gain would lie in $[5, 10]$ and that small gains would conflict with holding the formation; the sweep selected $\alpha^\star = 10$ and gave 0 filter-on GOAL outcomes with 1,251 INFEASIBLE at $\alpha = 0.5$ (Fig. 2). Because of the commit order these are documented, not pre-registered, predictions.

Remark (ZR). For these offset-point barriers the second-order sampled-data margin can be evaluated at the held command and written as two affine rows per barrier, which makes the zero command admissible whenever $h \geq 0$ in exact arithmetic (design document, Section 4a). Its ingredients are existing ideas [14, 17, 18], the inclusion of admissible sets is non-strict, and the property requires separate floating-point robustness analysis; it was not implemented or evaluated here.

Practical implications (suggestions, each tied to a result; not findings). (i) Assess numerical robustness at the boundary gain; investigate gains strictly below the boundary or an explicit tolerance rule in a separately designed follow-up (infeasibility audit). This is not evidence that a lower gain would perform better: in the development sweep lower gains reached GOAL less often (Fig. 2). (ii) Inspect the nominal policy's clearance distribution before deploying a filter: in the exploratory analysis the unfiltered minimum obstacle clearance separated pairs that lost completion under the filter (AUC 0.96). (iii) Include a brake baseline: B-match came within 1.0–3.2 GOAL points of the filter with no collisions or INFEASIBLE outcomes. (iv) The small unfiltered clearances in group A suggest clearance-aware or filter-aware training as a direction (exploratory).

9 Limitations

The study uses one simulated scenario, nine fixed actors from one custom MAPPO implementation, three obstacle densities and one gain. Validation layouts were reused for gain tuning. The formation criterion covers the common executed prefix only. Time to goal is analysed only in joint-success pairs. Safety statements are numerical under the design-document assumptions, not a continuous-time proof. The exploratory analyses are selected by observed outcome and descriptive. Three related-work classifications rest on abstracts only (Table 1).

10 Future work

Sensing noise or delay would test whether the unfiltered small-clearance passes stay collision-free. Clearance-aware or filter-aware retraining, and a gain strictly below the boundary or an explicit tolerance rule, are candidate follow-ups. Any such study would use the reserved, unused 5,000,000–5,999,999 layout block under its own locked protocol and disclose which of the present post-test findings informed it.

11 Reproducibility

Code, configuration, saved logs and analyses are in `https://github.com/mahdisallam860/cbf-marl-filter-study` (branch `claude/elegant-hopper-v8jxlv`). Protocol lock: commit `9baf6d7`; selected gain: `4fffe62`; final test: `e23f8c3` (per-step logs `41ae3d4`); mechanism analyses: `df78812`, `c2e331a`. Layout seeds: validation $2{,}000{,}000 + K$; final test $4{,}000{,}000 + 1{,}000p + K$ for actor index p; the block 5,000,000–5,999,999 is reserved and unused. The per-step logs of every final-test trial (poses, raw, clipped and applied commands, interventions, brake activations) are committed, so all analyses here can be repeated without re-running trials.

12 Mandatory disclosures

- The margins and protocol decisions were chosen by the author without advisor review; advisor agreement has not been obtained.
- The protocol was locked before any filter-on data with the frozen MAPPO actors existed, but after engineering stress tests with random and adversarial stand-in controllers and demonstration runs with an obstacle-blind stand-in controller (the locked document's wording "before any filter-on data existed" is corrected here).
- Filter-off validation results had been seen before the protocol was locked.
- The checkpoint rule was committed before the Phase 1 evaluation but after earlier validation results were seen.
- The validation layouts were reused as development data for gain tuning.
- The methodology report was committed after the final test.
- The mechanism categories were chosen after the confirmatory results were known.
- The post-test decision rule was not committed before the mechanism results were read.
- The locked protocol names the paired interval for criteria (a) and (b) "Newcombe method 10", but the analysis code implements method 8 (uncorrected correlation); with method 10 every bound changes by at most 0.0008 percentage points and no criterion changes (found after the final test).
- The example in Fig. 6 was first recorded as the unique nearest candidate; exact decimal comparison shows two equally close candidates, and the stated actor-index tie-break selects the same pair (regenerated metadata in `results/paper_revision/example_selection.json`).
- One of the 27,000 final-test trials (s2_early, $K = 3$, trial 646) started with a negative obstacle cover barrier ($h = -0.00777\,\mathrm{m}^2$, body clearance $0.087\,\mathrm{m}$), an exception to assumption A5; it is kept in all confirmatory numbers.
- Claims cover these nine fixed actors in this simulated scenario only.
- Safety statements are numerical under the design-document assumptions (A1–A9), not a continuous-time proof.
- The MAPPO implementation is custom and was sanity-checked only on the stock VMAS navigation task.

13 Conclusion

Under the pre-set criteria, the execution-time CBF-QP filter at $\alpha^\star = 10$ was not shown useful for these nine frozen MAPPO actors at any of the three densities: it reduced collision-or-infeasible failures but did not meet the non-inferiority criteria for goal completion and common-prefix formation error.

References

[1] L. Brunke, M. Greeff, A. W. Hall, Z. Yuan, S. Zhou, J. Panerati, A. P. Schoellig, "Safe learning in robotics: From learning-based control to safe reinforcement learning," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 5, pp. 411–444, 2022, doi:10.1146/annurev-control-042920-020211.

[2] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, P. Tabuada, "Control barrier functions: Theory and applications," Proc. 18th European Control Conference (ECC), 2019, pp. 3420–3431, doi:10.23919/ECC.2019.8796030.

[3] L. Wang, A. D. Ames, M. Egerstedt, "Safety barrier certificates for collisions-free multirobot systems," *IEEE Transactions on Robotics*, vol. 33, no. 3, pp. 661–674, 2017, doi:10.1109/TRO.2017.2659727.

[4] Z. Deng, Q. Li, M. Zhang, W. Chen, "Safe multi-agent reinforcement learning through neural graph control barrier functions," Proc. AAMAS 2026, pp. 3232–3234, extended abstract, doi:10.65109/KRTJ4225.

[5] M. Dawood, S. Pan, N. Dengler, S. Zhou, A. P. Schoellig, M. Bennewitz, "Safe multi-agent reinforcement learning for behavior-based cooperative navigation," *IEEE Robotics and Automation Letters*, vol. 10, no. 6, pp. 6256–6263, 2025, doi:10.1109/LRA.2025.3560830.

[6] Z. Cai, H. Cao, W. Lu, L. Zhang, H. Xiong, "Safe multi-agent reinforcement learning through decentralized multiple control barrier functions," arXiv:2103.12553, 2021.

[7] Z. Qin, K. Zhang, Y. Chen, J. Chen, C. Fan, "Learning safe multi-agent control with decentralized neural barrier certificates," ICLR 2021, arXiv:2101.05436.

[8] S. Zhang, O. So, M. Black, C. Fan, "Discrete GCBF proximal policy optimization for multi-agent safe optimal control," ICLR 2025, arXiv:2502.03640.

[9] J. J. Choi, J. J. Aloor, J. Li, M. G. Mendoza, H. Balakrishnan, C. Tomlin, "Resolving conflicting constraints in multi-agent reinforcement learning with layered safety," Robotics: Science and Systems (RSS), 2025, doi:10.15607/RSS.2025.XXI.094.

[10] S. Zhang, O. So, K. Garg, C. Fan, "GCBF+: A neural graph control barrier function framework for distributed safe multiagent control," *IEEE Transactions on Robotics*, vol. 41, pp. 1533–1552, 2025, doi:10.1109/TRO.2025.3530348.

[11] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," NeurIPS 2021, vol. 34, pp. 29304–29320, arXiv:2108.13264.

[12] A. P. Vinod, S. Safaoui, A. Chakrabarty, R. Quirynen, N. Yoshikawa, S. Di Cairano, "Safe multi-agent motion planning via filtered reinforcement learning," IEEE International Conference on Robotics and Automation (ICRA), 2022 (MERL TR2022-053). `https://www.merl.com/publications/docs/TR2022-053.pdf`

[13] A. Singletary, Y. Chen, A. D. Ames, "Control barrier functions for sampled-data systems with input delays," IEEE Conference on Decision and Control (CDC), 2020; arXiv:2005.06418.

[14] J. Breeden, K. Garg, D. Panagou, "Control barrier functions in sampled-data systems," *IEEE Control Systems Letters*, doi:10.1109/LCSYS.2021.3076127. `https://arxiv.org/abs/2103.03677` (Theorem 3, second-order approach; checked against arXiv v2 on 2026-10-26; the paper uses the opposite sign convention for h)

[15] X. Tan, E. Daş, A. D. Ames, J. W. Burdick, "Zero-order control barrier functions for sampled-data systems with state and input dependent safety constraints," arXiv:2411.17079, 2024.

[16] S. Liu, W. Xiao, C. A. Belta, "Sampling-aware control barrier functions for safety-critical and finite-time constrained control," arXiv:2511.11897, 2025.

[17] J. van den Berg, S. J. Guy, M. Lin, D. Manocha, "Reciprocal n-body collision avoidance," in *Robotics Research*, Springer Tracts in Advanced Robotics, vol. 70, Springer, 2011, doi:10.1007/978-3-642-19457-3_1.

[18] Y. Chen, A. Singletary, A. D. Ames, "Guaranteed obstacle avoidance for multi-robot operations with limited actuation: A control barrier function approach," *IEEE Control Systems Letters*, vol. 5, no. 1, 2021.

[19] Y. Emam, P. Glotfelter, M. Egerstedt, "Robust barrier functions for a fully autonomous, remotely accessible swarm-robotics testbed," arXiv:1909.02966, 2019.

[20] T. Kim, "Is your safe controller actually safe? A critical review of CBF tautologies and hidden assumptions," arXiv:2603.06954, 2026.

[21] G. Yang, C. Belta, R. Tron, "Self-triggered control for safety critical systems using control barrier functions," arXiv:1903.03692, 2019.

[22] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, Y. Wu, "The surprising effectiveness of PPO in cooperative multi-agent games," NeurIPS Datasets and Benchmarks Track, 2022; arXiv:2103.01955.

[23] M. Bettini, R. Kortvelesy, J. Blumenkamp, A. Prorok, "VMAS: A vectorized multi-agent simulator for collective robot learning," Proc. 16th International Symposium on Distributed Autonomous Robotic Systems (DARS), Springer, 2022.

[24] R. G. Newcombe, "Improved confidence intervals for the difference between binomial proportions based on paired data," *Statistics in Medicine*, vol. 17, pp. 2635–2650, 1998.